

Light-Phase Substrate Theory: Canonical Formulation as a Phase- First, Interference-First Substrate Theory with Emergent Geometry

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Abstract

We present the canonical formulation of Light-Phase Substrate Theory (LPST), a phase-first, interference-first substrate theory in which the sole fundamental object is a complex phase substrate $L = A e^{i\varphi}$ and the sole primitive operation is interference consistency. LPST is explicitly *not* an electromagnetic theory, *not* a mechanical wave model, *not* a medium-with-properties picture, and *not* a reinterpretation of spacetime. The word "light" in LPST refers to phase-bearing, interference-capable excitations and their maximal correlation-propagation structure — not to electromagnetic radiation. Space emerges as the metric representation of correlation decay between substrate labels: $d(x,y) = -\alpha \log C(x,y)$. Time emerges as the ordering of phase-update consistency events, with local proper time $d\tau \propto \sqrt{\kappa(x)} |\partial\varphi|$. Matter consists of topologically stabilized interference structures — phase vortices classified by $\pi_1(S^1) = \mathbb{Z}$, skyrmions classified by $\pi_2(S^2) = \mathbb{Z}$, and hopfions classified by $\pi_3(S^2) = \mathbb{Z}$ — whose stability is guaranteed by integer-valued topological charges. Energy and momentum are phase-gradient densities. Gravity emerges as the geometry induced by phase-stiffness gradients $\kappa(x) = \alpha A^2(x)$: topological defects suppress the local amplitude, creating stiffness gradients that modify the correlation kernel and thereby curve the emergent metric, recovering the linearized Schwarzschild solution and, via thermodynamic self-consistency, the full Einstein equations with higher-derivative corrections at Planckian curvature. Quantum mechanics is not added to the theory — it is native: amplitudes are fundamental,

interference is primitive, and the Feynman path integral emerges as a sum over phase-consistent histories. This canonical formulation locks the ontological primitives, defines the complete emergence stack (substrate $\rightarrow$ correlations $\rightarrow$ metric $\rightarrow$ causal structure $\rightarrow$ time $\rightarrow$ Lorentz symmetry $\rightarrow$ field dynamics), and establishes a strict boundary between LPST and electromagnetic field theory. Electromagnetism, if it emerges, does so as a sector of the low-energy effective theory — never as the foundation. LPST remains valid even in hypothetical universes with no electromagnetism.

Keywords: emergent spacetime, phase substrate, topological solitons, quantum gravity, interference, emergent geometry, correlation metrics, topological field theory

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1 Introduction — What LPST Is (and Is Not)

1.1 Motivation and Context

The two pillars of twentieth-century physics — quantum mechanics and general relativity — remain fundamentally incompatible at the level of their foundational assumptions. General relativity treats spacetime as a smooth, dynamical manifold whose curvature is determined by matter and energy through the Einstein field equations. Quantum mechanics, by contrast, operates on a fixed background and assigns probability amplitudes to observables defined on that background. The attempt to quantize gravity directly — treating the metric as a quantum field on a fixed topology — leads to a perturbatively non-renormalizable theory, signaling that the framework itself is inadequate [1, 2].

A growing body of evidence, both theoretical and heuristic, suggests that spacetime itself is not fundamental but emergent. The Bekenstein-Hawking entropy formula $S = A/(4G)$ associates entropy not with a volume of microstates but with an area, suggesting that the degrees of freedom underlying gravity are lower-dimensional [3, 4]. The AdS/CFT correspondence provides an explicit realization: a gravitational theory in $(d+1)$ -dimensional anti-de Sitter space is equivalent to a conformal field

theory in d dimensions with no gravity at all [5]. The Ryu-Takayanagi formula relates the entanglement entropy of boundary regions to the areas of minimal surfaces in the bulk [6], making the connection between entanglement and geometry precise. Jacobson's remarkable derivation of the Einstein equations from the Clausius relation $\delta Q = T dS$ applied to local Rindler horizons demonstrates that the field equations of general relativity can be regarded as an equation of state for spacetime thermodynamics [7]. The ER=EPR conjecture of Maldacena and Susskind posits a deep equivalence between Einstein-Rosen bridges and quantum entanglement [8]. Van Raamsdonk has argued that entanglement is the fundamental fabric holding spacetime together [9]. Verlinde has proposed that gravity is an entropic force arising from changes in the information content of the underlying microscopic theory [10].

All of these results point in the same direction: spacetime geometry is not a primitive structure but an emergent description of something deeper. What that deeper structure might be remains the central open question of theoretical physics. LPST proposes a specific and radical answer: the fundamental reality is a complex phase substrate whose sole dynamical principle is interference consistency, and all of physics — space, time, matter, energy, gravity, and quantum behavior — emerges from this substrate through a precisely defined hierarchy of constructions.

This paper presents the *canonical formulation* of LPST. The purpose of the canonical formulation is threefold: (i) to lock the ontological primitives of the theory so that no ambiguity remains about what LPST assumes and what it derives, (ii) to define the complete emergence stack — the precise sequence by which physical structures arise from the substrate, and (iii) to establish a strict boundary between LPST and electromagnetic field theory, thereby preventing the persistent misreading that conflates the substrate with Maxwell fields.

1.2 Negative Ontological Boundaries

Before stating what LPST is, we state what it is not. These negative boundaries are not rhetorical — they are essential to the logical integrity of the theory. Each negation prevents a specific category of confusion that has historically derailed similar proposals.

Definition 1.1 (Negative Ontological Boundaries).

LPST is explicitly not any of the following:

- (a) An electromagnetic theory;
- (b) A mechanical wave model;
- (c) A medium-with-properties picture;
- (d) A reinterpretation of spacetime.

(a) LPST is not an electromagnetic theory. The substrate field L is not a Maxwell field. It does not satisfy Maxwell's equations, it does not couple to charges via minimal coupling, and it does not transform as a connection on a $U(1)$ principal bundle in the standard sense of quantum electrodynamics. The word "light" in LPST does not refer to electromagnetic radiation. It refers to the phase-bearing, interference-capable character of the substrate and to the finite invariant speed at which correlations propagate. This distinction is critical: if LPST were merely a reformulation of electrodynamics, it would inherit the limitations of electrodynamics — linearity, conformal invariance in four dimensions, and the inability to generate massive particles without supplementary mechanisms. LPST is not so constrained. The electromagnetic sector, if it exists, is a low-energy emergent phenomenon, not the foundation.

The historical precedent for this confusion is instructive. In the late nineteenth century, the luminiferous ether was proposed as the medium through which electromagnetic waves propagate [11]. The ether was conceived as a mechanical substance filling all of space, with elastic properties that determined the speed of light. The Michelson-Morley experiment, the Lorentz-FitzGerald contraction hypothesis, and ultimately special relativity rendered the ether superfluous. The lesson is clear: identifying a fundamental substrate with the electromagnetic field leads to an over-determined and ultimately falsified theory. LPST avoids this by strict ontological separation.

(b) LPST is not a mechanical wave model. There is no medium vibrating in a pre-existing space. There are no elastic restoring forces, no mechanical dispersion relations, and no bulk properties such as density, pressure, or viscosity at the fundamental level. The substrate does not oscillate "in" space — it generates space through its correlation structure. A mechanical wave model presupposes exactly the structure that LPST derives: a background space in which waves propagate, and material properties that determine their speed. In LPST, there is no background space and no material properties. The phase field is not a displacement of a medium from equilibrium — it is the fundamental degree of freedom from which all other quantities are constructed.

This distinction separates LPST from analogue gravity models, which study sound waves in fluids as analogues of fields in curved spacetime [12]. Those models are instructive but fundamentally limited: the fluid exists in a pre-given spacetime, and the analogue geometry is an effective description superimposed on a more fundamental Newtonian or relativistic background. LPST makes no such concession. The emergence of geometry in LPST is not analogical but literal.

(c) LPST is not a medium-with-properties picture. The substrate is not an ether, not a superfluid, not a crystal lattice, and not a spin network in any conventional sense. It has no equation of state, no shear modulus, and no index of refraction at the fundamental level. The substrate has exactly two structural features: amplitude and phase. All other properties — including those that look like medium properties in certain limits — are derived.

This prevents confusion with Dirac-sea-like models [13] in which the vacuum is pictured as a filled Fermi sea of negative-energy states, or with more recent proposals involving superfluid vacuum theory [14] in which the vacuum is modeled as a Bose-Einstein condensate. While these models share the ambition of deriving geometry from something more fundamental, they all assume microscopic physics (quantum field theory, statistical mechanics) that LPST does not presuppose. LPST is prior to these frameworks.

(d) LPST is not a reinterpretation of spacetime. LPST does not reinterpret the notions of space and time by assigning them new meanings within an existing

framework. It *derives* space and time from constructs that do not presuppose them. This is the sharpest distinction: proposals that "reinterpret" spacetime (e.g., treating time as a parameter rather than a dimension, or treating space as relational rather than absolute) still operate within a framework where space and time are primitive notions. LPST operates at a level where neither space nor time exists — they must be constructed from the correlation structure and the phase-update ordering, respectively.

1.3 Positive Ontological Identity

Definition 1.2 (Positive Ontological Identity).

LPST is:

- (a) A phase-first, interference-first substrate theory;
- (b) A relational ontology in which only relations (phase differences, correlations) are physical;
- (c) A topological field theory with emergent geometry;
- (d) Compatible with electromagnetism but not reducible to it.

(a) Phase-first, interference-first. The phrase "phase-first" means that the complex phase $\varphi \in S^1$ of the substrate field is the carrier of all relational structure. It is the fundamental degree of freedom. The phrase "interference-first" means that the sole dynamical principle governing the substrate is the consistency of interference patterns — the requirement that the phase field be globally consistent across all closed loops. No Hamiltonian, no Lagrangian, and no time-evolution equation is assumed at the base level. All of these emerge from the interference-consistency constraint in appropriate limits. This is a radical departure from conventional physics, in which dynamics is typically specified by an action principle that presupposes a notion of time.

(b) Relational ontology. In LPST, no absolute quantities are physical. The absolute phase $\varphi(x)$ at a single substrate label x is unobservable — only phase differences $\Delta\varphi = \varphi(x) - \varphi(y)$ between labels are physically meaningful. Similarly, the absolute amplitude $A(x)$ is meaningful only through its role in defining the correlation kernel $C(x,y) = \langle L(x) L^*(y) \rangle$. This relational character is reminiscent of the philosophy underlying general relativity, where only relative measurements are physical, but in LPST it is pushed to a deeper level: even the labels x on which the substrate is defined have no intrinsic spatial or temporal meaning.

(c) Topological field theory with emergent geometry. The substrate supports topological defects — configurations with non-trivial winding numbers, skyrmion numbers, and Hopf invariants — that constitute matter. The dynamics of these defects is governed by the emergent geometry derived from the substrate's correlation structure. In this sense, LPST shares structural features with topological quantum field theories (TQFTs) [15, 16], but differs in that the geometry is itself emergent rather than a fixed background datum.

(d) Compatible with electromagnetism but not reducible to it. The substrate supports massless, transversely polarized propagating excitations in its vacuum sector. In a universe whose internal manifold M_{int} admits a $U(1)$ isometry, these excitations satisfy equations structurally identical to Maxwell's equations and can be identified with the electromagnetic field. But this identification is at the level of emergence: EM is a sector of the effective theory, not the substrate itself. If M_{int} had no $U(1)$ factor, there would be no electromagnetic sector, yet LPST would remain valid — space, time, matter, and gravity would still emerge. This is the definitive proof that LPST is not an electromagnetic theory.

1.4 The Fundamental Object

We state the foundational commitment of LPST in its most compressed form:

Fundamental Commitment

The fundamental object of LPST is a complex phase substrate whose only primitive operation is interference consistency. No other operation, property, or structure is assumed at the base level.

Every quantity in physics — distance, duration, mass, charge, force, curvature — is a derived quantity within LPST. The derivation proceeds through the emergence stack defined in Section 3. The remainder of this paper makes these derivations explicit.

2 Canonical LPST Primitives

The definitions in this section constitute the *locked* ontological core of LPST. They are non-negotiable: any formulation that modifies them is, by definition, not LPST. The locked status reflects the role of this section as the axiomatic foundation from which all subsequent results are derived.

2.1 The Substrate Field

Definition 2.1 (The Substrate Field).

The LPST substrate field is a map

$$L : \Sigma \rightarrow \mathbb{C} \times M_{\text{int}}$$

defined by

$$L(x) = A(x) e^{i\varphi(x)}, \quad u(x) \in M_{\text{int}} \quad (1)$$

where $A(x) \geq 0$ is the amplitude, $\varphi(x) \in S^1$ is the phase, $u(x)$ denotes optional internal degrees of freedom valued in a compact manifold M_{int} , and Σ is an abstract label space carrying no a priori geometric or temporal structure.

Several features of this definition require emphasis. First, the arguments $x \in \Sigma$ are abstract labels — they are *not* points in a pre-existing spacetime. The label space Σ is a set (possibly with a topology and a measure, but no metric) that indexes the degrees of freedom of the substrate. Coordinates, distances, and durations are emergent labels only, to be constructed from the correlation structure of L in Section 3. Any appearance of spatial or temporal arguments in intermediate calculations is a notational convenience, not a foundational assumption.

Second, the amplitude $A(x)$ is a real, non-negative scalar that sets the local stiffness and capacity for interaction of the substrate. It determines the local phase stiffness $\kappa(x) = \alpha A^2(x)$, which in turn controls the emergent geometry. Regions of suppressed amplitude correspond to regions of reduced phase stiffness, slower proper-time accumulation, and curved geometry — i.e., gravitational fields.

Third, the phase $\varphi(x)$ takes values in $S^1 \cong \mathbb{R}/2\pi\mathbb{Z}$ and is the carrier of all relational structure. The circle-valuedness of φ is essential: it permits topological defects (configurations where φ winds by $2\pi n$ around a closed loop) that constitute matter in LPST.

Fourth, the internal degrees of freedom $u(x)$ are optional in the minimal formulation but necessary for a realistic theory. They take values in a compact manifold M_{int} whose topology and isometry group determine the spectrum of topological defects (and hence the particle spectrum) and the gauge symmetries of the effective theory. If $M_{\text{int}} = S^2$, for instance, the theory supports skyrmions classified by $\pi_2(S^2) = \mathbb{Z}$.

Mathematically, L is a section of a fiber bundle over Σ with typical fiber $\mathbb{C} \times M_{\text{int}}$. The vacuum state is the trivial section:

$$L_{\text{vac}} = A_0 e^{i\varphi_0}, \quad u = u_0 = \text{const} \quad (2)$$

where A_0 , φ_0 , and u_0 are constants. The vacuum is the state of maximal uniformity: constant amplitude, constant phase, constant internal orientation. All non-trivial physics — matter, radiation, gravity — arises as deviations from this vacuum.

2.2 The Fundamental Rule

Axiom 1 (Phase Consistency Principle)

Allowed physical histories are those that minimize global phase inconsistency.

This axiom is the single dynamical principle of LPST. It replaces the role of the Hamiltonian, the Lagrangian, and the equations of motion in conventional physics. There is no time-evolution law at the base level of LPST. Dynamics *is* constraint satisfaction: the physical configuration of the substrate is determined by the requirement that the phase be globally consistent, up to topological obstructions.

To make this precise, we define the *phase inconsistency functional*:

$$\mathcal{I}[\varphi] = \sum_{\gamma} \oint_{\gamma} (\nabla\varphi \cdot d\mathbf{l} - 2\pi n_{\gamma}) \quad (3)$$

where the sum runs over all closed loops γ in the label space Σ , and $n_{\gamma} \in \mathbb{Z}$ is the winding number of φ around γ . The phase consistency principle states that physical configurations satisfy

$$\delta\mathcal{I} = 0 \quad (4)$$

for all variations $\delta\varphi$ that preserve the topological charges $\{n_{\gamma}\}$. Away from topological defects, this reduces to the flatness condition $\nabla \times \nabla\varphi = 0$, which is the statement that the phase field is locally exact (i.e., locally derivable from a single-valued function). At topological defects, the winding number is non-zero and the flatness condition is violated — but the violation is quantized: the circulation of $\nabla\varphi$ is exactly $2\pi n$ for integer n .

The phase consistency principle has deep structural parallels to several established principles in physics, each of which can be viewed as a special case or consequence:

(i) The *principle of stationary action* $\delta S = 0$ in classical mechanics presupposes a notion of time (the action is an integral over time). In LPST, the phase consistency

principle does not presuppose time — time emerges from the ordering of consistency events. The stationary-action principle is recovered in the temporal sector of the emergence stack (Section 3.3).

(ii) The *single-valuedness of the wavefunction* in quantum mechanics requires that the quantum phase be single-valued up to $2\pi n$ around any closed loop. This is precisely the content of Eq. (3) in the quantum-mechanical limit.

(iii) *Holonomy constraints* in gauge theory require that the parallel transport of a connection around a closed loop yield a group element determined by the enclosed flux. For the U(1) phase φ , this is again Eq. (3).

The unification of these three principles into a single axiom is one of the central achievements of LPST.

2.3 What "Light" Means in LPST

Definition 2.2 (Light in LPST).

In LPST, "light" does not mean electromagnetic radiation. "Light" refers to any excitation of the substrate that is:

- (i) Phase-bearing (carries a well-defined phase φ);*
- (ii) Interference-capable (participates in the phase consistency constraint);*
- (iii) Bounded in correlation propagation (propagates at a finite invariant speed);*
- (iv) Optionally endowed with polarization/helicity structure (from internal degrees u).*

This definition deliberately avoids conflation with Maxwell fields. The word "light" in LPST is used in the same spirit as "null" in general relativity — it refers to the structure of maximal correlation propagation, not to photons or electromagnetic waves. A null geodesic in GR is defined by the vanishing of the spacetime interval, $ds^2 = 0$, independently of whether the geodesic is traversed by a photon, a graviton,

or any other massless excitation. Similarly, "light" in LPST refers to the phase-propagation structure of the substrate, independently of whether that structure admits an electromagnetic interpretation.

The substrate would still be called a "light-phase substrate" even in a hypothetical universe with no electromagnetism. If the internal manifold M_{int} had no U(1) isometry, there would be no gauge-boson sector identifiable with photons, yet the substrate would still support phase-bearing, interference-capable excitations propagating at the invariant correlation speed. The name "LPST" is a statement about the substrate's structural properties — not about its relationship to the electromagnetic spectrum.

2.4 Notation and Conventions

Throughout this paper we adopt the following conventions unless otherwise stated:

- Natural units: $c = \hbar = 1$
 - Metric signature: $(-, +, +, +)$ for the emergent spacetime metric
 - Greek indices $\mu, \nu = 0, 1, 2, 3$ label emergent spacetime coordinates
 - Latin indices a, b label coordinates on the internal manifold M_{int}
 - Phase stiffness: $\kappa(x) = \alpha A^2(x)$
 - Correlation kernel: $C(x, y) = \langle L(x) L^*(y) \rangle$
 - Correlation length: ξ
 - Spatial correlation scale: ξ_s ; temporal correlation scale: ξ_t
 - Substrate coupling constants: $\alpha, \beta, \gamma, \lambda$
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3 The Emergence Stack

3.1 Overview of the Emergence Hierarchy

The central claim of LPST is that all of macroscopic physics emerges from the substrate through a precisely defined hierarchy. We present this hierarchy as an ordered sequence, where each level is constructed from the preceding levels and none is independently postulated:

Level	Structure	Constructed From	Section
0	Phase substrate L	Primitive (axiomatic)	§2
1	Correlations $C(x,y)$	Substrate field products	§3.2
2	Metric geometry $\mathcal{G}_{\mu\nu}$	Correlation decay	§3.2
3	Causal structure	Light-cone from correlations	§3.5
4	Time ordering	Phase-update consistency	§3.3
5	Lorentz symmetry	Vacuum correlation isotropy	§3.5
6	Field dynamics / particles	Topological defects + metric	§4, §6

The emergence stack is not merely heuristic — it is a logical dependency chain. Level n cannot be defined without Level $n-1$. Conversely, each level is fully determined by the levels below it, up to the choice of topological sector (i.e., the configuration of topological defects).

3.2 Emergence of Space

Definition 3.1 (Emergent Space).

Space is not a container. Space is a derived metric representation of correlation decay in the substrate.

The correlation kernel $C(x,y) = \langle L(x) L^*(y) \rangle$ encodes the statistical relationship between substrate degrees of freedom at distinct labels x and y . For the vacuum state, C is expected to decay with the "separation" of the labels, where separation is itself defined by the decay. The emergent distance is:

$$d^2(x,y) = -\ell^2 \ln(\langle C(x,y) \rangle / \langle C(x,x) \rangle) \quad (5)$$

or, in the simpler notation assuming normalized correlations:

$$d(x,y) = -\alpha \log C(x,y) \quad (6)$$

where ℓ is a fundamental length scale and α is a positive constant. This definition has a profound consequence: if $C(x,y) = 0$, the labels x and y have no spatial relation at all. They are not "infinitely far apart" — they simply do not participate in the same spatial manifold. Space exists only where correlations exist.

The emergent metric tensor is obtained by expanding Eq. (5) to second order in the separation $\Delta x^\mu = x^\mu - y^\mu$:

$$g_{\mu\nu}(x) = -(\ell^2/2) \partial_\mu \partial_\nu \ln |C(x,y)| \quad \text{evaluated at } y \rightarrow x \quad (7)$$

For a Gaussian correlation kernel in the vacuum, $C(x,y) = A_0^2 \exp(-|x-y|^2/2\xi^2)$, the emergent metric is flat:

$$g_{\mu\nu} = (\ell^2/\xi^2) \delta_{\mu\nu} \quad (8)$$

which is the Euclidean metric (up to a constant rescaling). For a Lorentzian correlation kernel with distinct spatial and temporal correlation scales ξ_s and ξ_t , the emergent metric acquires signature $(-, +, +, +)$, yielding the Minkowski metric of special relativity. More generally, when topological defects modify the local

correlation structure, the emergent metric acquires non-trivial curvature — this is the origin of gravity in LPST (Section 5).

The correlation-to-metric construction has deep parallels to the holographic prescriptions of AdS/CFT. The Ryu-Takayanagi formula [6] relates entanglement entropy in the boundary CFT to areas of minimal surfaces in the bulk, implying that bulk geometry is encoded in boundary correlations. Van Raamsdonk's observation [9] that reducing entanglement entropy between two regions corresponds to increasing the distance between them is precisely the LPST mechanism of Eq. (5): less correlation implies greater emergent distance. The key difference is that LPST does not require a holographic boundary or an asymptotically AdS bulk — the construction is intrinsic to the substrate.

3.3 Emergence of Time

Definition 3.2 (Emergent Time).

Time is not a dimension. Time is the ordering of phase-update consistency events in the substrate.

In LPST, there is no global time parameter at the fundamental level. What we experience as time is the sequential ordering imposed by the phase consistency principle: a configuration at "time t_2 " is the unique phase-consistent continuation of the configuration at "time t_1 ," given the topological boundary conditions. The ordering is partial (not total), which is consistent with the relativity of simultaneity in special and general relativity.

The local proper time is defined as:

$$d\tau \propto \sqrt{\kappa(x)} \cdot |\partial\varphi| \quad (9)$$

where $\kappa(x) = \alpha A^2(x)$ is the local phase stiffness. This formula has an immediate physical consequence: *time dilation is the redistribution of the phase-update budget due to local substrate conditions.* Near a massive topological defect, the amplitude

$A(x)$ is suppressed (the defect "drains" amplitude from its neighborhood), so $\kappa(x)$ decreases, and proper time accumulates more slowly. This is gravitational time dilation.

For a static, spherically symmetric defect of mass M , the amplitude profile is (Section 5):

$$A(r) \approx A_0(1 - GM/(rc^2)) \quad (10)$$

The ratio of proper times at two radial positions is:

$$d\tau_1/d\tau_2 = A(x_1)/A(x_2) \approx \sqrt{(1 - 2GM/r)} \quad (11)$$

which is precisely the Schwarzschild time dilation factor. This is not an analogy — it is a derivation.

The LPST treatment of time dissolves the notorious "problem of time" in quantum gravity [17, 18]. In canonical quantum gravity (Wheeler-DeWitt theory), the Hamiltonian constraint $H|\Psi\rangle = 0$ implies that the wavefunction of the universe is static, leading to the paradox that nothing evolves. The Page-Wootters mechanism [19] resolves this by treating time as an internal correlation between subsystems. LPST implements a similar resolution: there is no external time, but the internal phase-update ordering provides a relational notion of change. The Wheeler-DeWitt equation is recovered as the constraint that the total phase inconsistency vanishes.

3.4 Emergence of Energy and Momentum

Definition 3.3 (Energy and Momentum).

Energy is the phase-gradient density. Momentum is directional phase flow. Both are bookkeeping quantities derived from the phase structure — not independent primitives.

The substrate action, written in terms of the emergent coordinates, takes the form:

$$S[L, u] = \int d^4x \sqrt{(-g)} [\alpha(\partial_\mu A)(\partial^\mu A) + \alpha A^2(\partial_\mu \varphi)(\partial^\mu \varphi) + \beta A^2(\partial_\mu u^a)(\partial^\mu u_a) + \gamma K - \lambda(A^2 - A_0^2)^2] \quad (12)$$

where K is the topological stabilizer (Skyrme term, discussed in Section 4.5) and the last term is a potential that stabilizes the vacuum amplitude. The energy-momentum tensor derived from this action is:

$$T_{\mu\nu} = \alpha[(\partial_\mu A)(\partial_\nu A) + A^2(\partial_\mu \varphi)(\partial_\nu \varphi)] + \beta A^2(\partial_\mu u^a)(\partial_\nu u_a) - g_{\mu\nu} \mathcal{L} + \gamma T_{\mu\nu}^{(K)} \quad (13)$$

The mass-energy equivalence $E = mc^2$ is derived, not postulated. Consider a topological defect at rest with rest mass M (defined in Section 4.4). Under a Lorentz boost with velocity v , the phase gradients transform according to the emergent Lorentz symmetry (Section 3.5), yielding:

$$E = \gamma M c^2, \quad p = \gamma M v \quad (14)$$

where $\gamma = (1 - v^2/c^2)^{-1/2}$. This is the standard relativistic energy-momentum relation, derived from the transformation properties of the substrate phase gradients.

3.5 Emergence of Lorentz Invariance

Lorentz invariance in LPST is an emergent symmetry of the vacuum, not a postulated symmetry of the fundamental theory. In the vacuum state, the correlation kernel depends on the label separation only through the combination:

$$C_{\text{vac}}(x, y) = f(\eta_{\mu\nu} \Delta x^\mu \Delta x^\nu) \quad (15)$$

where $\eta_{\mu\nu} = \text{diag}(-\xi_t^{-2}, \xi_s^{-2}, \xi_s^{-2}, \xi_s^{-2})$ and f is a monotonically decreasing function. The emergent metric (7) then inherits the $\text{SO}(3,1)$ symmetry of the argument of f , reproducing the Lorentz group.

The emergent speed of light is the ratio of spatial to temporal correlation scales:

$$c = \xi_s / \xi_t \quad (16)$$

This identification gives c a clear substrate-level meaning: it is not the speed of any substance, but the invariant ratio that determines the structure of the emergent light cone.

Because Lorentz invariance is emergent, LPST predicts that it is violated at sufficiently high energies — specifically, at energies approaching the substrate's fundamental scale E_p . The predicted dispersion relation for high-energy excitations is:

$$v(E) = 1 - \eta(E/E_p)^n \quad (17)$$

where η is a dimensionless coefficient of order unity and $n \geq 1$ depends on the detailed structure of the substrate at ultra-short correlation lengths. For $n = 1$, current constraints from Fermi-LAT observations of gamma-ray bursts place $E_p > 10^{19}$ GeV, consistent with the Planck scale [20]. For $n = 2$, the constraints are weaker: $E_p > 10^{11}$ GeV. LPST naturally accommodates $n = 2$ with E_p at the Planck scale, yielding effects at the threshold of current observational sensitivity — a testable prediction.

4 Matter in LPST

4.1 What Matter Is Not

Remark 4.1.

Matter is NOT "trapped light." This phrasing implies a mechanical picture — photons bouncing inside a box, standing electromagnetic waves confined by reflective walls — that is completely foreign to LPST. There are no photons at the substrate level, no walls, no trapping. The phrase "trapped light" conflates the substrate with the electromagnetic field and reintroduces precisely the medium-with-properties picture that Definition 1.1 excludes.

The "trapped light" picture fails on multiple grounds. First, it presupposes the existence of photons, which are emergent objects in LPST (if they exist at all — their existence depends on the structure of M_{int}). Second, it presupposes a spatial box or cavity, which requires pre-existing space — exactly what LPST derives rather than assumes. Third, it suggests that matter is unstable (the "photons" could eventually escape), whereas topological stability in LPST is exact. Fourth, it implies that matter has no properties other than those of its constituent "photons," whereas the topological structure of defects gives rise to a rich spectrum of quantum numbers (charge, baryon number, spin) that have no counterpart in the trapped-light picture.

4.2 What Matter Is

Definition 4.1 (Matter).

Matter is topologically stabilized interference structure. It consists of localized, finite-energy configurations of the substrate that cannot be continuously deformed to the vacuum. Their stability is guaranteed by topology — integer-valued topological charges that are exactly conserved under any continuous deformation of the fields.

The key word is *topologically*. A topological defect is a field configuration in which the fields, evaluated on a sphere S^n surrounding the defect, define a map $S^n \rightarrow M_{\text{target}}$ that is not contractible — it belongs to a non-trivial element of the homotopy group $\pi_n(M_{\text{target}})$. Because homotopy classes are discrete (they are labeled by integers or elements of a discrete group), the defect cannot be continuously deformed away. It is stable for topological reasons, regardless of energetic considerations.

This is the sense in which matter in LPST is "interference structure": a defect is a pattern in the phase and polarization fields that, due to its topological winding, cannot be unwound. It is stabilized not by a confining potential or by external forces,

but by the mathematical impossibility of continuously interpolating between distinct homotopy classes.

4.3 Classification of Matter

The allowed matter classes in LPST are determined by the homotopy groups of the target spaces of the substrate fields. We present the complete classification:

(a) Phase vortices. These are line-like defects (in 3+1 dimensions) or point-like defects (in 2+1 dimensions) around which the phase φ winds by $2\pi n$:

$$\oint_V \nabla \varphi \cdot d\mathbf{l} = 2\pi n, \quad n \in \mathbb{Z} \quad (18)$$

They are classified by $\pi_1(S^1) = \mathbb{Z}$. The winding number n is identified with electric charge (in units of a fundamental charge determined by the substrate coupling constants). The energy of a vortex diverges logarithmically with system size for an isolated vortex in two dimensions and linearly in three dimensions — but vortex-antivortex pairs have finite energy. The identification of phase winding with electric charge is motivated by the structural identity between the phase-consistency constraint (3) and the Gauss law constraint in electrostatics.

(b) Texture solitons / Skyrmions. These are localized configurations of the internal field $u: \mathbb{R}^3 \rightarrow S^2$ that wrap the target sphere a non-trivial number of times:

$$B = (1/4\pi) \int d^2x \quad u \cdot (\partial_1 u \times \partial_2 u) \in \mathbb{Z} \quad (19)$$

They are classified by $\pi_2(S^2) = \mathbb{Z}$. The integer B is the baryon number [21, 22]. Skyrmions have the remarkable property that a 2π rotation induces a sign change in the field configuration, endowing them with *fermionic statistics* — a result first demonstrated by Finkelstein and Rubinstein [23]. This provides a substrate-level explanation for why baryons are fermions: their fermionic character is a topological consequence, not an additional postulate.

(c) Hopfions (knots). These are knotted solitons classified by the Hopf invariant, involving maps $S^3 \rightarrow S^2$:

$$H = (1/4\pi^2) \int d^3x \quad \varepsilon^{ijk} F_{ij} A_k \in \mathbb{Z} \quad (20)$$

where F_{ij} is the pull-back of the area form on S^2 and A_k is the corresponding Chern-Simons potential. They are classified by $\pi_3(S^2) = \mathbb{Z}$. The preimages of distinct points on S^2 are closed curves in $\mathbb{R}^3$ that are linked — and this linking prevents the soliton from being unwound. Hopfions provide a candidate for additional topological quantum numbers beyond charge and baryon number.

(d) Braids (worldline topology). In 2+1 dimensions, the exchange of identical particles is governed not by the permutation group S_n but by the braid group B_n [24]. This allows for *anyonic statistics* — particles whose exchange phase is neither 0 (bosons) nor π (fermions) but an arbitrary angle θ . While 3+1 dimensions restrict statistics to bosonic and fermionic, the braid-group structure is relevant to LPST in the context of topological quantum computation and effective 2+1-dimensional systems.

We summarize the homotopy classification in the following table:

Homotopy Group	Classification	Defect Type	Physical Charge
$\pi_1(S^1) = \mathbb{Z}$	Winding number n	Phase vortex	Electric charge
$\pi_2(S^2) = \mathbb{Z}$	Skyrmion number B	Skyrmion / texture	Baryon number
$\pi_3(S^2) = \mathbb{Z}$	Hopf invariant H	Hopfion / knot	Linking charge

4.4 Mass and Inertia

The rest mass of a topological defect is defined as the total phase-gradient energy in the rest frame:

$$M = \int d^3x [\alpha A_0^2 (\nabla \varphi)^2 + \beta A_0^2 (\nabla u)^2 + \gamma K + \lambda (A^2 - A_0^2)^2] \quad (21)$$

Each term has a clear physical meaning: the first is the energy stored in phase gradients, the second in polarization gradients, the third is the topological stabilization energy, and the fourth is the amplitude-restoring energy. The mass is

finite for any topological defect of finite winding number, provided the stabilizer K is included (see Section 4.5).

Definition 4.2 (Inertia).

Inertia is the cost of re-phasing an extended topological object. When a defect is accelerated, the entire phase and polarization profile must be coherently re-established in the new frame. The resistance to this re-phasing is inertia.

This definition avoids any mechanical picture of "stuff resisting motion." There is no stuff, and there is no pre-existing notion of motion at the substrate level. What exists is a topological field configuration that, by virtue of its spatial extent, requires a coordinated rearrangement of phase and polarization over a finite region when its state of motion (defined via the emergent Lorentz group) is changed. The larger and more tightly wound the defect, the more costly the re-phasing — hence the greater the inertia. This provides a substrate-level explanation for why mass is additive (to leading order): the re-phasing cost of a composite system is the sum of the re-phasing costs of its constituents, plus interaction terms.

4.5 Evasion of Derrick's Theorem

Derrick's theorem [25] states that static, finite-energy solutions of scalar field theories with only standard kinetic terms do not exist in more than one spatial dimension. The proof proceeds by a scaling argument: under the rescaling $x \rightarrow \lambda x$, the gradient energy scales as λ^{2-d} while the potential energy scales as λ^{-d} , and no extremum of the total energy exists for $d \geq 2$.

LPST evades Derrick's theorem through the topological stabilizer K , which is a higher-derivative term of the Skyrme type [21]:

$$K = [(\partial_\mu u \times \partial_\nu u) \cdot (\partial^\mu u \times \partial^\nu u)] \quad (22)$$

Under the rescaling $x \rightarrow \lambda x$, the Skyrme term energy scales as λ^{4-d} , which in $d = 3$ gives λ^1 . The total energy then has the schematic form:

$$E(\lambda) = E_2/\lambda + E_4\lambda \quad (23)$$

where E_2 is the standard kinetic energy contribution (scaling as λ^{-1} in $d = 3$) and E_4 is the Skyrme term contribution (scaling as λ^{+1}). The minimum occurs at $\lambda_* = \sqrt{(E_2/E_4)}$, giving a soliton of finite size $R \sim \sqrt{(\gamma/\alpha)}$. This is the mechanism by which LPST defects acquire definite size and finite mass.

4.6 Defect Interactions and Charge Conservation

Topological defects interact through the overlap of their phase and polarization profiles. For two phase vortices with winding numbers n_1 and n_2 , separated by a distance r in the emergent metric, the interaction energy is:

$$V(r) = -2\pi\alpha A_0^2 n_1 n_2 \ln(r\xi) \quad (2+1 \text{ dim.}) \quad (24)$$

$$V(r) \propto n_1 n_2 / r \quad (3+1 \text{ dim.}) \quad (25)$$

In 3+1 dimensions, this is precisely the Coulomb potential. Vortices of the same sign ($n_1 n_2 > 0$) repel; vortices of opposite sign attract. The coupling constant is determined by the substrate parameters: $e^2 = 4\pi\alpha A_0^2$.

Charge conservation in LPST follows from *topological necessity*, not from Noether's theorem. The total winding number of a configuration is a topological invariant — it cannot change under continuous evolution of the fields. The total charge can only change by the creation or annihilation of vortex-antivortex pairs, which have zero net winding number. This is a stronger conservation law than Noether's theorem provides: it holds exactly, non-perturbatively, and without reference to any symmetry of an action functional. It is a consequence of the mathematical structure of $\pi_1(S^1) = \mathbb{Z}$.

5 Gravity in LPST

5.1 What Gravity Is Not

Remark 5.1.

Gravity is NOT refraction through a medium. This phrasing implies a pre-existing medium with bulk optical properties — an index of refraction, a density profile, a mechanical response. LPST has no medium. The substrate is not a material through which "light rays" are bent; it is the fundamental structure from which geometry itself arises. The refraction analogy is dangerous because it suggests that gravity is a force acting on pre-existing trajectories, whereas in LPST gravity is a modification of the metric structure that defines what trajectories are.

5.2 What Gravity Is

Definition 5.1 (Gravity).

Gravity is emergent geometry caused by phase-stiffness gradients. Topological defects suppress the local amplitude $A(x)$, creating gradients in the phase stiffness $\kappa(x) = \alpha A^2(x)$. These gradients modify the correlation kernel $C(x,y)$, which modifies the emergent metric $g_{\mu\nu}$ via the correlation-to-metric map (7). Objects follow geodesics of the correlation geometry — they are not pushed by forces.

The mechanism is as follows. A topological defect of mass M has an amplitude profile that deviates from the vacuum value A_0 . In the neighborhood of the defect, the phase consistency constraint requires that A be suppressed — the non-trivial phase

winding "costs" amplitude. The resulting profile $A(x)$ determines the local phase stiffness $\kappa(x) = \alpha A^2(x)$, which enters the correlation kernel:

$$C(x, y) \propto \exp(-|x - y|^2 / 2\xi^2(x)) \quad (26)$$

where the correlation length $\xi(x)$ depends on the local stiffness. The correlation-to-metric map (7) then yields a non-flat emergent metric — a curved spacetime. Other defects (test particles) follow geodesics of this metric, producing the phenomenon we call gravitational attraction.

5.3 The Effective Metric

The effective metric in the presence of a phase-stiffness gradient can be written as:

$$g^{\mu\nu}_{\text{eff}} \sim \kappa^{-1} \eta^{\mu\nu} + \Delta^{\mu\nu} \quad (27)$$

where $\eta^{\mu\nu}$ is the background (vacuum) Minkowski metric and $\Delta^{\mu\nu}$ encodes corrections from defect profiles. This structure is reminiscent of the Gordon metric [26]:

$$g^{\mu\nu}_{\text{Gordon}} = \eta^{\mu\nu} + (1 - n^{-2}) u^\mu u^\nu \quad (28)$$

which describes the effective geometry experienced by light in a dielectric medium with refractive index n and four-velocity u^μ . The resemblance is structural, not accidental — but the critical difference is that in LPST, the Gordon-like metric (27) is *fundamental* rather than an analogy for a medium effect. There is no underlying flat metric on which a dielectric sits; the effective metric *is* the geometry.

This structural parallel was noted in the context of analogue gravity by Unruh [12], Visser [27], and others, who showed that sound waves in a moving fluid experience an effective metric. LPST takes this insight to its logical conclusion: if the metric *is* the correlation structure of the substrate, then the "analogy" between medium effects and gravity is not an analogy — it is an identity, at the emergent level.

5.4 Recovery of Newtonian Gravity

For a static, spherically symmetric defect of mass M , the amplitude profile at distances r much larger than the defect core size is:

$$A(r) \approx A_0(1 - GM/(rc^2)) \quad (29)$$

where G is a constant determined by substrate parameters. The phase stiffness is:

$$\kappa(r) = \alpha A_0^2(1 - 2GM/(rc^2) + \dots) \quad (30)$$

Substituting into the correlation-to-metric map, the emergent metric becomes:

$$ds^2 \approx -(1 - 2GM/r)dt^2 + (1 + 2GM/r)(dx^2 + dy^2 + dz^2) \quad (31)$$

This is the linearized Schwarzschild metric in isotropic coordinates. Geodesic motion in this metric yields Newton's law of gravitation:

$$d^2\mathbf{x}/dt^2 = -\nabla\Phi, \quad \Phi = -GM/r \quad (32)$$

Newton's gravitational constant is determined by substrate parameters:

$$G \propto 1/(\alpha A_0^2) \quad (33)$$

The physical meaning is transparent: a stiffer substrate (larger αA_0^2) produces weaker gravitational effects, because the same mass defect produces a smaller relative perturbation in the stiffness.

5.5 Recovery of General Relativity

The full Einstein equations are recovered from two requirements:

(i) *Self-consistency*. The metric appearing in the substrate action (12) must equal the metric derived from the correlations of the substrate via Eq. (7). This is a bootstrap condition: the geometry determines the dynamics of the substrate, and the dynamics of the substrate determines the geometry. The self-consistency requirement is a set of equations relating the metric $g_{\mu\nu}$ to the energy-momentum tensor $T_{\mu\nu}$.

(ii) *Thermodynamic derivation*. Following Jacobson [7], we apply the Clausius relation $\delta Q = T dS$ to local Rindler horizons in the substrate. The heat flux δQ is the energy flux of the substrate through the horizon, the temperature T is the Unruh temperature $T = a/(2\pi)$, and the entropy is proportional to the horizon area: $dS = \eta dA$. Demanding that this relation hold for all local Rindler observers yields:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu} \quad (34)$$

which are the Einstein field equations with a cosmological constant Λ [7]. In LPST, the proportionality constant η — and hence Newton's constant G — is determined by the substrate parameters through Eq. (33). The cosmological constant arises as the vacuum energy of the substrate, $\Lambda = \lambda(A_0^2 - A_0^2)^2$ evaluated with radiative corrections, which is a version of the cosmological constant problem that LPST inherits but does not yet resolve (see Section 8.2).

5.6 Corrections Beyond GR

The self-consistency condition in LPST determines the emergent gravitational dynamics not only at leading order (Einstein equations) but also at subleading orders, producing higher-derivative corrections:

$$S_{\text{grav}} = \int d^4x \sqrt{(-g)} [R/(16\pi G) + c_1 R^2 + c_2 R_{\mu\nu} R^{\mu\nu} + c_3 R_{\mu\nu\rho\sigma} R^{\mu\nu\rho\sigma} + \dots] \quad (35)$$

where the coefficients c_i are of order ξ^2 (the square of the correlation length). These corrections are entirely negligible at laboratory scales and astrophysical scales, where the curvature radius is vastly larger than ξ . However, they become significant in two regimes:

(i) *Near singularities.* As the curvature approaches $R \sim 1/\xi^2$, the higher-derivative terms become comparable to the Einstein-Hilbert term, modifying the singularity structure. In particular, the Schwarzschild singularity at $r = 0$ is expected to be resolved — replaced by a smooth, high-curvature core where the substrate amplitude A approaches zero but remains finite.

(ii) *In the early universe.* During the Planck epoch, the curvature of the universe approaches the substrate scale, and the higher-derivative corrections dominate. These corrections may modify the initial conditions for inflation, provide a natural mechanism for a bounce cosmology, or alter the spectrum of primordial perturbations — all of which are potentially observable in the cosmic microwave background [28].

6 Quantum Behavior as LPST-Native

6.1 Quantum Mechanics Is Not Added

Proposition 6.1.

Quantum mechanics is not an additional postulate in LPST. It emerges naturally because:

- (i) Amplitudes are fundamental — the substrate IS a complex amplitude field;*
- (ii) Interference is primitive — it is the sole dynamical principle;*
- (iii) Topology is discrete — topological charges are integer-valued, providing exact quantization of observable quantities.*

This proposition is the key insight distinguishing LPST from classical field theories that must be "quantized" by imposing commutation relations or by constructing a path integral over field configurations. In LPST, the path integral and the complex amplitude structure are already present at the substrate level — they do not need to be imposed from outside. The substrate *is* a quantum object, in the precise sense that its dynamics is governed by interference of complex amplitudes.

The standard axioms of quantum mechanics — the superposition principle, the Born rule, the projection postulate — are all derived within LPST rather than postulated. The following subsections make these derivations explicit.

6.2 Superposition

Definition 6.1 (Superposition).

Superposition is the coexistence of consistent phase histories. Multiple phase configurations can satisfy the phase consistency principle simultaneously. The

substrate naturally supports their coherent coexistence.

In conventional quantum mechanics, the superposition principle is a postulate: if $|\psi_1\rangle$ and $|\psi_2\rangle$ are allowed states, then $\alpha|\psi_1\rangle + \beta|\psi_2\rangle$ is also an allowed state. In LPST, this follows from the linearity of the phase consistency constraint in the small-amplitude regime. When the phase field admits multiple solutions to the consistency equation (4), all solutions coexist — the substrate does not "choose" one. The relative amplitudes and phases of the coexisting solutions determine the interference pattern, which in turn determines the probabilities of measurement outcomes.

Superposition is not a mysterious quantum postulate in LPST — it is the default state of an interference-based substrate. The mystery, if any, is not why superpositions exist, but why they appear to collapse upon measurement. LPST addresses this in Section 6.4.

6.3 The Path Integral

The phase consistency principle directly yields the Feynman path integral [29]. The transition amplitude between an initial substrate configuration L_i and a final configuration L_f is:

$$\langle L_f | L_i \rangle = \int [DL] \exp(iS[L]/\hbar) \quad (36)$$

where the sum (functional integral) runs over all phase-consistent configurations interpolating between L_i and L_f , and $S[L]$ is the substrate action (12). The weight $\exp(iS/\hbar)$ is the substrate's native assignment of complex amplitude to each history — it is not imposed by a quantization procedure.

Planck's constant $\hbar$ is determined by the fundamental scale of phase quantization:

$$\hbar \sim \alpha A_0^2 \xi^2 \quad (37)$$

This gives $\hbar$ a substrate-level meaning: it is the product of the vacuum stiffness and the squared correlation length. A stiffer substrate with a shorter correlation length

gives a smaller $\hbar$ — and hence more "classical" behavior. The classical limit $\hbar \rightarrow 0$ corresponds to $\alpha A_0^2 \xi^2 \rightarrow 0$ with the macroscopic quantities held fixed, i.e., to a limit where the substrate's fundamental scale becomes infinitely fine-grained compared to the scales of interest.

6.4 Measurement

Definition 6.2 (Measurement).

Measurement is a local phase-locking event. When a topological defect (observer/apparatus) becomes phase-correlated with another defect (system), the measurement interaction establishes definite phase relations between them. This is decoherence — not collapse.

The measurement interaction proceeds as follows. Before the measurement, the system defect is in a superposition of phase configurations, and the apparatus defect is in a definite ("ready") state. The interaction brings the two defects into phase correlation: the apparatus acquires a phase configuration that is entangled with the system's phase configuration. After the interaction, the reduced density matrix of the system — obtained by tracing over the apparatus degrees of freedom — is diagonal in the measurement basis. The off-diagonal elements (coherences) decay because the apparatus has a vast number of internal degrees of freedom (u), and the phase information is distributed among all of them. This is standard decoherence theory [30], but in LPST it is not an *ad hoc* addition — it follows from the substrate dynamics.

The decoherence timescale is:

$$\tau_{\text{dec}} \sim \hbar / (E_{\text{int}} \cdot N_{\text{eff}}) \quad (38)$$

where E_{int} is the interaction energy and N_{eff} is the effective number of apparatus degrees of freedom. For macroscopic apparatuses, $N_{\text{eff}} \sim 10^{23}$, and decoherence is essentially instantaneous.

6.5 Probability and the Born Rule

Definition 6.3 (Probability).

Probability is the relative volume of consistent phase-space. Among all phase-consistent configurations, destructive interference suppresses random-phase contributions and constructive interference enhances phase-aligned ones.

The Born rule $P(a) = |\psi_a|^2$ follows from a phase-space volume argument. Consider the set of all substrate histories consistent with a measurement outcome a . Each history contributes a complex amplitude $\exp(iS/\hbar)$. For an outcome a with amplitude ψ_a , the histories cluster around the stationary-phase point, and the volume of the cluster (in the Gaussian approximation around the saddle point) is proportional to $|\psi_a|^2$. This is the phase-space derivation of the Born rule, closely related to Zurek's envariance argument [31] and to Gleason's theorem in the finite-dimensional case [32].

6.6 The Schrödinger Equation

The Schrödinger equation is derived as the non-relativistic limit of the substrate dynamics for a defect of mass M in a slowly varying potential $V(x)$. Writing the substrate field in the WKB form $L = A(x, t) \exp(iS(x, t)/\hbar)$, substituting into the substrate equation of motion, and keeping terms to leading order in $\hbar$, we obtain the Hamilton-Jacobi equation. At next order, we obtain the continuity equation for A^2 . Combining both — defining $\psi(x, t) = A(x, t) \exp(iS(x, t)/\hbar)$ — we recover:

$$i\hbar \partial\psi/\partial t = -(\hbar^2/2M)\nabla^2\psi + V(x)\psi \quad (39)$$

This is the Schrödinger equation. In LPST, it is not a fundamental equation — it is the non-relativistic, single-defect limit of the substrate dynamics, valid when the defect velocity is much smaller than the invariant correlation speed and when the external potential varies slowly compared to the defect's internal structure.

6.7 No Collapse Postulate Required

We emphasize that LPST requires no collapse postulate, no many-worlds interpretation [33], and no hidden variables [34]. The appearance of definite measurement outcomes is explained by decoherence (Section 6.4), which is a consequence of the unitary substrate dynamics acting on macroscopic systems with many internal degrees of freedom. The "measurement problem" — the question of how and when the quantum state "collapses" into a definite outcome — is dissolved: there is no collapse. The substrate evolves unitarily at all times. What appears as collapse is the loss of coherence in the reduced density matrix of the measured system, caused by its entanglement with the apparatus and, ultimately, with the environment.

This resolution is not unique to LPST — it is shared by decoherence-based interpretations of quantum mechanics. What is unique to LPST is that the decoherence mechanism is not *added* to the theory — it is *built into* the substrate dynamics from the start. The substrate is a complex amplitude field whose dynamics is governed by interference. Decoherence — the suppression of interference between macroscopically distinct histories — is an automatic consequence of this dynamics when the number of degrees of freedom is large.

7 LPST vs. LST — Boundary Closure

7.1 The Conceptual Drift Problem

The strict boundary between LPST and any formulation that treats the electromagnetic field as foundational is not a matter of taste or emphasis — it is a logical necessity. Without this boundary, LPST risks being misread as the claim that "everything is made of electromagnetic waves," which is:

- Physically wrong (EM is a spin-1 gauge field; the substrate is not);
- Historically regressive (it revives the luminiferous ether in new language);

- Empirically falsifiable in ways that LPST is not (e.g., a universe without EM would falsify "everything is EM" but not LPST);
- Technically limiting (it restricts the emergence of non-abelian gauge symmetries, massive particles, and gravitational degrees of freedom).

The boundary must be established with maximum clarity. We therefore dedicate this section to a systematic disambiguation between LPST (the canonical frame) and what we call "LST" (Light Substrate Theory) — any formulation that, intentionally or through conceptual drift, treats the electromagnetic field as the substrate.

7.2 Three Non-Assumptions

Axiom 2 (Non-Assumptions)

(i) LPST does not assume electromagnetic fields. The substrate is prior to and more fundamental than any specific gauge sector.

(ii) LPST does not assume mechanical propagation. There is no wave equation in a pre-existing medium. The substrate does not "propagate through" anything — it generates the stage on which propagation is defined.

(iii) LPST does not privilege c as a substance speed. The invariant $c = \xi_s/\xi_t$ is the correlation-propagation bound — the ratio of spatial to temporal correlation scales in the vacuum. It is not the speed of any stuff.

Each of these non-assumptions prevents a specific interpretive error. Non-assumption (i) prevents the identification L = electromagnetic potential, which would reduce LPST to a nonlinear electrodynamics. Non-assumption (ii) prevents the introduction of a background space through which waves travel, which would undermine the emergence of space from correlations. Non-assumption (iii) prevents the identification of c with the speed of photons, which would tie the invariant speed

to a specific particle species rather than to the structure of the vacuum correlation kernel.

7.3 If EM Emerges, It Does So as a Sector

The vacuum of the substrate supports small-amplitude, long-wavelength excitations of the phase field φ . In the linearized regime around the vacuum $\varphi = \varphi_0 + \delta\varphi$, the phase consistency principle reduces to:

$$\nabla^2 \delta\varphi = 0 \quad (\text{static}), \quad \square \delta\varphi = 0 \quad (\text{dynamic}) \quad (40)$$

The dynamic equation is the massless wave equation. Its solutions propagate at the invariant speed c and have two transverse polarization degrees of freedom (arising from the two independent orientations of $\nabla \delta\varphi$ perpendicular to the propagation direction). These properties — masslessness, transversality, two polarizations, propagation at c — are precisely those of the electromagnetic field in the Coulomb gauge.

It is therefore natural to *identify* these excitations with photons and the linearized phase dynamics with Maxwell's equations. But this identification is at the level of emergence: the photon is a low-energy, long-wavelength excitation of the phase sector of the substrate. It is not the substrate itself.

The full gauge structure of the Standard Model — $G_{\text{SM}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$ — must similarly emerge from the isometries of the internal manifold M_{int} . The electromagnetic U(1) is just one factor. The non-abelian factors SU(3) and SU(2) arise from the non-abelian isometries of M_{int} . The specific structure of M_{int} that yields exactly G_{SM} is an open problem (Section 8.2), but the *mechanism* — gauge symmetry from isometry — is well-established in Kaluza-Klein theory [35, 36] and its generalizations.

7.4 The Universe-Independence Test

Proposition 7.1 (Universe-Independence).

LPST remains valid even in hypothetical universes with no electromagnetism.

Consider a hypothetical universe in which the internal manifold M_{int} has no U(1) isometry — for instance, $M_{\text{int}} = \text{SU}(2)/\mathbb{Z}_3$, a space whose isometry group is SU(2) with no U(1) factor. In this universe:

- The substrate $L = A e^{i\phi}$ still exists.
- The phase consistency principle still holds.
- Correlations still decay, generating an emergent spatial metric.
- Phase-update ordering still generates emergent time.
- Topological defects (skyrmions in $M_{\text{int}} = \text{SU}(2)/\mathbb{Z}_3$) still exist and constitute matter.
- Phase-stiffness gradients around defects still curve the emergent metric, producing gravity.
- Quantum behavior still emerges from the interference of complex amplitudes.

The only difference is that there is no massless, transversely polarized excitation identifiable as a photon, and no U(1) gauge symmetry identifiable as electromagnetism. This universe has space, time, matter, gravity, and quantum mechanics — but no light, in the electromagnetic sense. Yet the substrate is still a "light-phase substrate" in the LPST sense: it still has phase, it still interferes, and its correlations still propagate at a finite invariant speed.

This thought experiment is the definitive demonstration that LPST is not an electromagnetic theory. Any formulation that would fail in this hypothetical universe is not LPST — it is something else.

7.5 Comparison Table

Aspect	LST (Incorrect Reading)	LPST (Canonical Frame)
Fundamental entity	Electromagnetic field	Complex phase substrate $L = Ae^{i\varphi}$
Role of EM	Foundation — everything is EM	Emergent sector — one factor of the low-energy gauge group
Nature of "light"	Electromagnetic radiation (photons)	Phase-bearing, interference-capable excitation
Role of c	Speed of EM waves in vacuum	Invariant correlation-propagation bound ξ_s/ξ_t
Matter model	Trapped EM waves	Topologically stabilized interference structures
Gravity model	Refraction of EM in a medium	Emergent geometry from phase-stiffness gradients
Quantum behavior	Added via quantization of EM	Native — substrate IS quantum
Background assumptions	Pre-existing spacetime, Maxwell equations	None — all structure emergent
Valid without EM?	No	Yes (Proposition 7.1)

7.6 Summary: The Cleanest Boundary

The ontological boundary between LPST and electromagnetic field theory is the defining feature of the canonical LPST frame. It can be summarized in a single sentence:

Any formulation that treats the electromagnetic field as foundational rather than emergent is not LPST.

This boundary is not negotiable, not approximate, and not a matter of convention. It is a logical consequence of the axioms of Section 2. The substrate is defined without reference to Maxwell's equations. The dynamics is defined without reference to electromagnetic coupling. The emergence of space, time, matter, and gravity proceeds without any electromagnetic input. If electromagnetism appears, it appears as a consequence — a sector of the low-energy effective theory determined by the isometry structure of $\mathcal{M}_{\text{int}}$. This is the canonical frame.

8 Discussion and Outlook

8.1 Summary of the Canonical Frame

We have presented the canonical formulation of Light-Phase Substrate Theory. The theory is built on a minimal axiomatic foundation:

- **One substrate:** $L = A e^{i\varphi}$ with optional internal degrees $u \in \mathcal{M}_{\text{int}}$
- **One rule:** Allowed physical histories minimize global phase inconsistency (Axiom 1)
- **Everything emergent:** Space (correlation decay), time (phase-update ordering), matter (topological defects), energy/momentum (phase gradients), gravity (stiffness gradients), quantum behavior (native interference), Lorentz invariance (vacuum correlation symmetry), gauge symmetry (internal manifold isometry)

The canonical frame locks the ontological primitives, defines the emergence stack, and draws a strict boundary against electromagnetic reductionism. No additional postulates are required. No background structures are assumed. The theory is maximally minimal: it posits the least possible structure and derives the most possible physics.

8.2 Open Problems

Despite the conceptual completeness of the canonical frame, several problems remain open. These are not defects of the theory — they are research directions that the theory generates.

(i) Determination of M_{int} . The internal manifold must be chosen so that its isometry group is exactly the Standard Model gauge group $G_{\text{SM}} = \text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$. The Kaluza-Klein mechanism [35, 36] requires M_{int} to have dimension ≥ 7 (for $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$), but the specific choice of manifold — and the mechanism by which it is selected over other possibilities — remains undetermined. Candidate manifolds include coset spaces such as $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1) / (\text{SU}(2) \times \text{U}(1) \times \text{U}(1))$ [37], as well as more exotic constructions from algebraic topology.

(ii) Fine-structure constant. The fine-structure constant $\alpha_{\text{EM}} \approx 1/137$ must emerge from the substrate coupling constants $\alpha, \beta, \gamma, \lambda$ and the geometric properties of M_{int} . Computing this from first principles is a key quantitative test of the theory.

(iii) Cosmological constant problem. The vacuum energy of the substrate generically gives a cosmological constant Λ of order $1/\xi^2$, which is $\sim 10^{122}$ times larger than the observed value. This is the standard cosmological constant problem [38], which LPST inherits. A resolution may involve a dynamical mechanism that drives the vacuum amplitude A_0 to a value that nearly cancels the bare vacuum energy, or a topological mechanism that selects a vacuum sector with anomalously small Λ .

(iv) Black hole information. LPST provides a natural framework for addressing the black hole information paradox [39]. The substrate has no event horizon at the fundamental level — horizons are emergent features of the effective metric (31). At the substrate level, information is encoded in the phase and polarization configuration and is never destroyed. The unitarity of substrate evolution guarantees that information is preserved, but the detailed mechanism by which it escapes the emergent horizon remains to be worked out.

(v) Lattice simulation. The substrate action (12) is amenable to lattice simulation using Monte Carlo methods. Numerical studies of the phase structure, the defect

spectrum, the emergent metric in the presence of defects, and the approach to the continuum limit are all within reach of current computational resources.

(vi) Observational signatures. LPST makes several predictions that are in principle testable:

- Lorentz violation at ultra-high energies: $v(E) = 1 - \eta(E/E_P)^n$ (Eq. 17), constrainable by Fermi-LAT [20] and future gamma-ray observatories
- Gravitational wave dispersion: frequency-dependent propagation speed for gravitational waves, constrainable by LIGO/Virgo [40]
- CMB anomalies: modifications to the primordial power spectrum from higher-derivative corrections (35) during inflation, constrainable by Planck and future CMB experiments [28]
- Topological defect signatures: cosmic strings from phase vortices, domain walls from polarization textures, potentially observable in gravitational wave backgrounds

8.3 Relation to Other Approaches

LPST occupies a unique position in the landscape of quantum gravity approaches. We briefly compare it to the major programs:

Loop quantum gravity (LQG) [41, 42] discretizes spacetime at the Planck scale using spin networks and spin foams. Like LPST, LQG does not assume a background spacetime. Unlike LPST, LQG starts from the canonical quantization of general relativity — it is a "quantize-first" approach, whereas LPST is a "substrate-first" approach in which both gravity and quantum mechanics emerge simultaneously from the phase consistency principle.

String theory [43, 44] posits one-dimensional extended objects (strings) as fundamental. Strings naturally include gravity (the graviton appears in the closed-string spectrum) and gauge symmetry (from compactification or brane constructions). However, string theory requires a background spacetime for its perturbative formulation and does not provide a background-independent definition

at the non-perturbative level (M-theory remains conjectural). LPST is background-independent by construction.

Causal set theory [45] replaces the spacetime manifold with a discrete partial order — a locally finite set with a causal relation. The continuum emerges in the limit of large sets. Like LPST, causal set theory takes a "spacetime is emergent" stance. Unlike LPST, it privileges causal structure as fundamental, whereas in LPST causal structure emerges from the correlation kernel.

Causal dynamical triangulations (CDT) [46] constructs spacetime from simplicial building blocks glued together with a causal constraint. CDT has successfully reproduced the large-scale dimensionality of spacetime and provides a well-defined path integral for quantum gravity. LPST differs in that its fundamental degrees of freedom are continuous (a field on an abstract label space), not discrete (simplices).

Penrose twistor theory [47] encodes spacetime geometry in the complex-analytic structure of twistor space, where points in spacetime correspond to lines in twistor space. The primacy of complex structure in twistor theory resonates with LPST's primacy of the complex phase field. A possible connection — in which the substrate label space Σ is identified with (a generalization of) twistor space — is a direction for future investigation.

Jacobson thermodynamic gravity [7] derives the Einstein equations from thermodynamic relations applied to local Rindler horizons. LPST incorporates this derivation as a step in its emergence stack (Section 5.5), giving it a microscopic foundation: the entropy is the phase-space volume of substrate configurations consistent with the macroscopic horizon data.

Wheeler's "it from bit" [48] proposes that physics is fundamentally informational. LPST is consistent with this philosophy — information in LPST is encoded in phase relations, and the phase consistency principle is an information-theoretic constraint. But LPST goes further by specifying the concrete substrate that carries the information.

The distinctive feature of LPST in this landscape is its combination of properties: (i) continuous field, (ii) no background, (iii) emergent everything (space, time, gravity, QM, gauge symmetry), (iv) concrete dynamical principle (phase consistency), (v) explicit matter model (topological defects), and (vi) testable predictions (Lorentz violation, gravitational wave dispersion). No other approach combines all six.

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